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1 Data Management SS26: Exercise 01 – Data Modeling

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- This exercise on data modeling aims to provide practical experience in Entity-Relationship (ER) modeling, ER-relational mapping, and relational normalization.
- Please upload the required files of the sub-tasks individually to TeachCenter - *do NOT compress several files together using, e.g., `.zip`*. This means we expect exactly one `.pdf` file and one `.sql` file for this assignment.
- IMPORTANT: This exercise must be completed individually. Any form of plagiarism, copying, or unauthorized collaboration will be sanctioned according to academic regulations. Strict measures will be taken against any violations.

1.1 ER Modeling (15/25 points)

Create an ER diagram in Modified Chen (MC) notation – including entity types, relationship types, attributes, cardinalities, and keys (create surrogate keys if natural unique identifiers are missing) – for managing data stored on Ecommerce platform. Please, use the digital tool **draw.io**[1] for designing the ER diagram. We recommend **draw.io** since it has a toolbar for ER elements (among others). There are multiple correct ways of modeling (alternatives, and level of details), however the diagram should capture the essential information of the following discourse:

- The store manages information about *products*. Each *product* is described by a unique *product_id*. Additionally, the database stores the *product name*, *brand*, *category*, *color*, *size*, *gender*, *cost price*, and *catalog price* (list price). A *product* may exist in the database without appearing in any *sale*.
- The store manages inventory through *stock*. To uniquely identify a *stock* entry, the system requires both the product and its storage country. This ensures that inventory quantities are managed separately for each location. The database stores the *stock quantity* available for each product in each specific country. A single *product* may have multiple *stock* entries across different countries, while each *stock* entry corresponds to exactly one *product*.
- The store records *sales* (orders). Each *sale* is uniquely identified by a *sale_id*. For each *sale*, the database records the *sale date*, the *country*, and a *discounted* indicator (specifying if a discount was applied). The *total amount* of the transaction is calculated from the associated sale items, shipping costs, and discounts. Each *sale* is associated with exactly one *customer*, while a *customer* may be associated with multiple *sales*. It is important to note that a *customer* may exist in the database without having any *sales* recorded.

[1] `https://app.diagrams.net/`

- Each *sale* consists of one or more *sale items*. Each *sale item* is identified by a unique *item_id*. This entity links a *sale* to a *product* and stores detailed transaction data, including the *quantity*, the *original price*, the *unit price* (price actually paid), the *discount applied* (absolute value), the *discount percent*, a *discounted* indicator, and the final *item total*. Each *sale item* refers to exactly one *sale* and exactly one *product*.

- The database contains data on *customers* registered in the store. Each *customer* is identified by a unique *customer_id*. Additionally, the database stores the customer's *country*, *age range*, and *signup date*.

- *Sales* can be influenced by promotional *campaigns*. Each *campaign* is uniquely identified by a *campaign_id*. The database stores the *campaign name*, the *channel* (e.g., Email, Social Media), *start date*, *end date*, *discount type* (percentage or fixed amount), and *discount value*. A *sale* can be associated with at most one *campaign*, while a *campaign* may be associated with many *sales*. A *sale* may reference a *campaign* only if the *sale date* lies within the campaign period. Furthermore, *campaigns* may target specific *products*. A single *campaign* can promote multiple *products*, and a product can be associated with multiple *campaigns*.

  Please restrict yourself to using the following annotations for cardinalities: `M, MC, C, 1`.

Deliverable Sub-Task 1.1: `ERDiagram.pdf` (1 page PDF)

1.2 Mapping ER Diagrams into the Relational Model (10/25 points)

Create a relational schema for the ER diagram designed in Task 1.1 and bring it into third normal form. This schema should include the relations and typed attributes, as well as all primary and foreign keys. Then, use the **PostgreSQL**[2],[3]Data Base Management System to create a SQL script (`CreateSchema.sql`) with `CREATE TABLE` statements.

Deliverable Sub-Task 1.2: `CreateSchema.sql`

[2] `https://www.postgresql.org/download/`
[3] `https://tc.tugraz.at/main/mod/resource/view.php?id=480253`